

Student Segmentation and Personalized Course Recommendation System for EduPro

Abstract

Online learning platforms serve students with different learning goals, skill levels, and interests, but many still provide the same course recommendations to every learner. This project presents a data-driven Student Segmentation and Personalized Course Recommendation System for EduPro. It analyzes learner demographics, enrollment history, course preferences, ratings, and spending behavior to create meaningful learner profiles. Using feature engineering and K-Means clustering, students are grouped into behavioral segments such as beginners, specialized learners, and career-focused learners. Based on each segment, the system recommends relevant courses using content-based filtering and learner preference analysis. An interactive Streamlit dashboard allows users to explore learner profiles, view enrolled courses, and receive personalized learning paths. The proposed solution improves course discovery, enhances learner engagement, and supports adaptive, student-centric learning experiences, making online education more personalized and effective.

1. Introduction

Online learning has transformed education by making high-quality courses accessible to learners anytime and anywhere. However, students have different learning goals, backgrounds, and interests. Some learners explore multiple subjects, while others focus on a specific domain or pursue career-oriented certifications. As a result, providing the same course recommendations to every user often leads to poor engagement and reduced learning satisfaction.

This project presents a Student Segmentation and Personalized Course Recommendation System for EduPro. The system analyzes learner demographics, enrollment history, course categories, ratings, and spending behavior to understand individual learning patterns. Using data analytics and K-Means clustering, learners are grouped into meaningful behavioral segments. Based on these segments, personalized course recommendations are generated to help students discover relevant learning paths.

The proposed solution aims to improve learner engagement, increase course completion, and support a more adaptive and student-centric online learning experience.

2. Problem Statement

EduPro is an online learning platform that offers courses across multiple domains and skill levels. However, its current recommendation approach follows a one-size-fits-all model, where learners receive similar course suggestions regardless of their interests or learning behavior. This makes it difficult for students to discover courses that match their goals and abilities.

The main problems addressed in this project are:

- Lack of personalized course recommendations.
- Limited understanding of learner behavior and preferences.
- Low learner engagement and course completion.
- No structured framework for student segmentation.

To solve these challenges, this project builds a data-driven learner segmentation model and generates personalized course recommendations using behavioral analysis and machine learning, helping EduPro deliver a more engaging and student-centric learning experience.

3. Dataset Description

The EduPro project uses three datasets to analyze learner behavior and generate personalized course recommendations. These datasets contain demographic, course, and transaction information for each learner.

Dataset Overview

Dataset	Description	Key Fields
Users	Stores learner demographic details	UserID, Age, Gender
Courses	Contains course information	CourseID, CourseCategory, CourseType, CourseLevel, CourseRating
Transactions	Records course enrollment and payments	UserID, CourseID, TransactionDate, Amount

Dataset Details

The Users dataset identifies each learner through a unique UserID and includes demographic attributes such as age and gender.

The Courses dataset describes every course offered on the platform, including its category, type, difficulty level, and average learner rating.

The Transactions dataset connects learners with the courses they enrolled in. It records the enrollment date and the amount paid, making it possible to analyze learner engagement and spending behavior.

After data cleaning, the three datasets are merged using UserID and CourseID to create a single analytical dataset for feature engineering, clustering, and personalized recommendations.

4. Methodology

The project follows a structured data analytics workflow to understand learner behavior and generate personalized course recommendations. First, data is collected from the Users, Courses, and Transactions datasets. The data is then cleaned by removing missing values and ensuring consistent formats. All three datasets are merged using UserID and CourseID to create a unified learner database. Feature engineering is performed to generate meaningful learner attributes such as total enrollments, spending behavior, preferred category, and diversity score. Next, numerical features are standardized and K-Means clustering is applied to segment learners into behavioral groups. Finally, a personalized recommendation system suggests relevant courses based on learner preferences and cluster behavior. The complete solution is presented through an interactive Streamlit dashboard.

5. Feature Engineering

Feature engineering converts raw enrollment data into learner-level insights. For every student, several behavioral features are calculated, including the total number of courses enrolled, average spending, average course rating, and diversity score representing the number of categories explored. The system also identifies each learner's preferred course category and preferred course level using enrollment frequency. In addition, a learning depth index measures whether a learner focuses on beginner or advanced courses. These engineered features create comprehensive learner profiles that improve the accuracy of clustering and personalized recommendations.

6. Results

The proposed system successfully segments learners into meaningful behavioral groups using K-Means clustering. Students with similar learning patterns are grouped together based on enrollment history, spending, and course preferences. The recommendation engine provides relevant courses by considering each learner's preferred category and difficulty level. The Streamlit dashboard demonstrates personalized learning paths, making course discovery easier and improving the overall learning experience. The results show that behavioral segmentation can support higher engagement and more effective personalized education.

7. Streamlit Dashboard

An interactive Streamlit dashboard was developed to make the analysis accessible and user-friendly. Users can select any learner through a dropdown menu and instantly view demographic information, enrolled courses, and personalized recommendations. The dashboard displays learner profiles in a clear tabular format and supports interactive exploration of course categories and learning levels. This web application transforms analytical results into a practical decision-support tool for both learners and educational platforms.

8. Conclusion

This project demonstrates how data analytics and machine learning can improve online learning through student-centric personalization. By combining learner demographics, behavioral features, and clustering techniques, EduPro can better understand different learner types and recommend suitable courses. The interactive Streamlit dashboard makes the system practical, scalable, and easy to use. Overall, the solution enhances learner engagement, supports adaptive learning journeys, and provides a strong foundation for future AI-based educational recommendation systems.

References

1. Scikit-learn Documentation
2. Pandas Documentation
3. Streamlit Documentation
4. K-Means Clustering Research